

Module 11: Inheritance — Mendelian & Polygenic Genetics — Study Questions

Mendel's Laws

1. What organism did Mendel study? Why was it a good choice for genetics experiments?
2. Define the Law of Segregation. How does it relate to the separation of homologs during Meiosis I?
3. Define the Law of Independent Assortment. Does the inheritance of one trait affect another (for unlinked genes)?
4. What is a test cross? How can it determine whether an organism with a dominant phenotype is homozygous or heterozygous?

Monohybrid Crosses

1. Explain the difference between homozygous dominant (AA), heterozygous (Aa), and homozygous recessive (aa).
2. If "T" = Tall and "t" = Short, what is the phenotype of "Tt"? What is the genotype called?
3. Cross $Tt \times Tt$. What are the expected genotypic and phenotypic ratios?
4. Cross $Tt \times tt$ (a test cross). What are the expected phenotypic ratios? What would the results tell you about the Tt parent?

Dihybrid Crosses

1. What phenotypic ratio is expected from a dihybrid cross ($AaBb \times AaBb$) assuming independent assortment?

2. In peas, Yellow (Y) is dominant to green (y), and Round (R) is dominant to wrinkled (r).
Cross $YyRr \times YyRr$. How many of the 16 offspring boxes in the Punnett square will show yellow, round peas?

Beyond Simple Dominance

1. Differentiate between incomplete dominance and codominance. Give an example of each.
2. In snapdragons, red (RR) $\times$ white (rr) produces all pink (Rr). Cross two pink flowers ($Rr \times Rr$). What are the expected phenotypic ratios?
3. Explain the ABO blood type system. What alleles are involved? Which are codominant? Which is recessive?
4. Can a Type A mother ($I^A I^A$ or $I^A i$) and a Type B father ($I^B I^B$ or $I^B i$) have a Type O child? Show the Punnett square for the case where this is possible.
5. What is pleiotropy? How does the sickle cell allele illustrate this concept?

Sex-Linked Inheritance

1. Why are males more frequently affected by X-linked recessive disorders (e.g., color blindness, hemophilia)?
2. A carrier (heterozygous) mother and an unaffected father have children. What fraction of their sons will be affected? What fraction of their daughters will be carriers?
3. Can a father pass an X-linked trait to his sons? Why or why not?

Pedigree Analysis

1. Two unaffected parents have an affected child. Is the trait most likely dominant or recessive? Explain.
2. In a pedigree, affected individuals appear in every generation. Is the trait most likely dominant or recessive?

3. A trait appears almost exclusively in males and is never passed from father to son. What inheritance pattern does this suggest?

Polygenic Inheritance

1. What is a polygenic trait? How does the inheritance of height differ from the inheritance of pea plant height (tall vs. short)?
2. Why do polygenic traits show continuous variation instead of distinct categories?
3. If skin color is controlled by three genes, each with two alleles (Aa, Bb, Cc), how many phenotypic categories are possible? What shape does the distribution form?
4. A student says: "If I know someone's genotype, I know exactly what they'll look like." Is this accurate? How does the environment modify gene expression? Give a specific example.

Integration and Application

1. Hydrangea flower color changes with soil pH. Does genotype always strictly determine phenotype? Discuss the interplay of genes and environment.
2. Explain how Mendel's Law of Segregation connects to the physical process of homolog separation during Meiosis I ([link to Module 9](#)).
3. Compare single-gene inheritance (Mendelian) and polygenic inheritance using a table with: number of genes, phenotype categories, example traits, and distribution pattern.
4. Two parents who are both heterozygous for a recessive genetic disorder ($Aa \times Aa$) want to know the probability of having an affected child. Calculate the probability, and explain why genetic counseling might use Punnett squares.
5. Explain why studying human genetics is more challenging than studying pea genetics. What tools (pedigrees, population data, molecular testing) compensate for the inability to perform controlled crosses?